



AR 506: Cognitive Robotics

Course Code	AR 506
Course Name	Cognitive Robotics
L-T-P-C	3-0-0-3
Prerequisites	Consent of faculty advisor
Intended for	UG, PG and PhD
Mutual Exclusion	None
Approval	50th BoA (Jul 2023)

Course Contents

- **Introduction:** Introduction to Cognitive Robotics and Human-Robot Interaction, Smart Materials. (6 hours)
- **Brain physiology and neural signal transmission:** Architecture of the Brain, Nerve cells, Synchronization Models, Electroencephalography. (6 hours)
- **Intelligence architecture:** Theories of Intelligence, Kuramoto Model, Child-Robot Interaction. (10 hours)
- **Artificial cognitive systems:** Cognitive architectures, The CRAM cognitive architecture. (10 hours)
- **Functional programming:** Robot manipulation and task-level robot programming using ROS, The CRAM plan language. (10 hours)

Text Books

1. Cangelosi A. and Asada M., **Cognitive Robotics**, The MIT Press, 2022.
2. Samani H., **Cognitive Robotics**, CRC Press, 2015.

References

1. Purves D., et al., **Neuroscience**, Sinauer Associates, 2004.
2. Pfeifer R. and Bongard J., **How the body shapes the way we think-A New View of Intelligence**, MIT Press.
3. Raol J. R., and Ayyagari R., **Control Systems: Classical, Modern, and AI-Based Approaches**, CRC Press.